

```
In [1]: import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [3]: income_df = pd.read_csv(r"C:\Users\sayee\Downloads\Inc_Exp_Data (1).csv")
```

```
In [4]: income_df.head()
```

```
Out[4]:
```

|   | Mthly_HH_Income | Mthly_HH_Expense | No_of_Fly_Members | Emi_or_Rent_Amt | Annua |
|---|-----------------|------------------|-------------------|-----------------|-------|
| 0 | 5000            | 8000             | 3                 | 2000            |       |
| 1 | 6000            | 7000             | 2                 | 3000            |       |
| 2 | 10000           | 4500             | 2                 | 0               |       |
| 3 | 10000           | 2000             | 1                 | 0               |       |
| 4 | 12500           | 12000            | 2                 | 3000            |       |

## analyze the data

```
In [5]: income_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                       50 non-null     int64
1   Mthly_HH_Expense                     50 non-null     int64
2   No_of_Fly_Members                    50 non-null     int64
3   Emi_or_Rent_Amt                      50 non-null     int64
4   Annual_HH_Income                     50 non-null     int64
5   Highest_Qualified_Member              50 non-null     object
6   No_of_Earning_Members                 50 non-null     int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB
```

```
In [6]: income_df.shape
```

```
Out[6]: (50, 7)
```

```
In [21]: income_df.columns
```

```
Out[21]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
               'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
               'No_of_Earning_Members'],
              dtype='object')
```

```
In [22]: len(income_df.columns)
```

```
Out[22]: 7
```

In [23]: `income_df.describe()`

Out[23]:

|              | Mthly_HH_Income | Mthly_HH_Expense | No_of_Fly_Members | Emi_or_Rent_Amt | Ar |
|--------------|-----------------|------------------|-------------------|-----------------|----|
| <b>count</b> | 50.000000       | 50.000000        | 50.000000         | 50.000000       |    |
| <b>mean</b>  | 41558.000000    | 18818.000000     | 4.060000          | 3060.000000     |    |
| <b>std</b>   | 26097.908979    | 12090.216824     | 1.517382          | 6241.434948     |    |
| <b>min</b>   | 5000.000000     | 2000.000000      | 1.000000          | 0.000000        |    |
| <b>25%</b>   | 23550.000000    | 10000.000000     | 3.000000          | 0.000000        |    |
| <b>50%</b>   | 35000.000000    | 15500.000000     | 4.000000          | 0.000000        |    |
| <b>75%</b>   | 50375.000000    | 25000.000000     | 5.000000          | 3500.000000     |    |
| <b>max</b>   | 100000.000000   | 50000.000000     | 7.000000          | 35000.000000    |    |

In [7]: `income_df.describe().T`

Out[7]:

|                              | count | mean      | std           | min     | 25%      | 50%      |
|------------------------------|-------|-----------|---------------|---------|----------|----------|
| <b>Mthly_HH_Income</b>       | 50.0  | 41558.00  | 26097.908979  | 5000.0  | 23550.0  | 35000.0  |
| <b>Mthly_HH_Expense</b>      | 50.0  | 18818.00  | 12090.216824  | 2000.0  | 10000.0  | 15500.0  |
| <b>No_of_Fly_Members</b>     | 50.0  | 4.06      | 1.517382      | 1.0     | 3.0      | 4.0      |
| <b>Emi_or_Rent_Amt</b>       | 50.0  | 3060.00   | 6241.434948   | 0.0     | 0.0      | 0.0      |
| <b>Annual_HH_Income</b>      | 50.0  | 490019.04 | 320135.792123 | 64200.0 | 258750.0 | 447420.0 |
| <b>No_of_Earning_Members</b> | 50.0  | 1.46      | 0.734291      | 1.0     | 1.0      | 1.0      |

In [9]: `income_df.isna().any()`

Out[9]:

```

Mthly_HH_Income      False
Mthly_HH_Expense      False
No_of_Fly_Members     False
Emi_or_Rent_Amt       False
Annual_HH_Income      False
Highest_Qualified_Member  False
No_of_Earning_Members  False
dtype: bool

income_df.isna()

```

In [10]: `income_df["Mthly_HH_Expense"].mean()`

Out[10]: `np.float64(18818.0)`

In [24]: `income_df.isna()`

Out[24]:

|    | Mthly_HH_Income | Mthly_HH_Expense | No_of_Fly_Members | Emi_or_Rent_Amt | Annu |
|----|-----------------|------------------|-------------------|-----------------|------|
| 0  | False           | False            | False             | False           |      |
| 1  | False           | False            | False             | False           |      |
| 2  | False           | False            | False             | False           |      |
| 3  | False           | False            | False             | False           |      |
| 4  | False           | False            | False             | False           |      |
| 5  | False           | False            | False             | False           |      |
| 6  | False           | False            | False             | False           |      |
| 7  | False           | False            | False             | False           |      |
| 8  | False           | False            | False             | False           |      |
| 9  | False           | False            | False             | False           |      |
| 10 | False           | False            | False             | False           |      |
| 11 | False           | False            | False             | False           |      |
| 12 | False           | False            | False             | False           |      |
| 13 | False           | False            | False             | False           |      |
| 14 | False           | False            | False             | False           |      |
| 15 | False           | False            | False             | False           |      |
| 16 | False           | False            | False             | False           |      |
| 17 | False           | False            | False             | False           |      |
| 18 | False           | False            | False             | False           |      |
| 19 | False           | False            | False             | False           |      |
| 20 | False           | False            | False             | False           |      |
| 21 | False           | False            | False             | False           |      |
| 22 | False           | False            | False             | False           |      |
| 23 | False           | False            | False             | False           |      |
| 24 | False           | False            | False             | False           |      |
| 25 | False           | False            | False             | False           |      |
| 26 | False           | False            | False             | False           |      |
| 27 | False           | False            | False             | False           |      |
| 28 | False           | False            | False             | False           |      |
| 29 | False           | False            | False             | False           |      |
| 30 | False           | False            | False             | False           |      |
| 31 | False           | False            | False             | False           |      |
| 32 | False           | False            | False             | False           |      |

|    | Mthly_HH_Income | Mthly_HH_Expense | No_of_Fly_Members | Emi_or_Rent_Amt | Annu |
|----|-----------------|------------------|-------------------|-----------------|------|
| 33 | False           | False            | False             | False           |      |
| 34 | False           | False            | False             | False           |      |
| 35 | False           | False            | False             | False           |      |
| 36 | False           | False            | False             | False           |      |
| 37 | False           | False            | False             | False           |      |
| 38 | False           | False            | False             | False           |      |
| 39 | False           | False            | False             | False           |      |
| 40 | False           | False            | False             | False           |      |
| 41 | False           | False            | False             | False           |      |
| 42 | False           | False            | False             | False           |      |
| 43 | False           | False            | False             | False           |      |
| 44 | False           | False            | False             | False           |      |
| 45 | False           | False            | False             | False           |      |
| 46 | False           | False            | False             | False           |      |
| 47 | False           | False            | False             | False           |      |
| 48 | False           | False            | False             | False           |      |
| 49 | False           | False            | False             | False           |      |

income\_df.columns

In [25]: income\_df.columns

Out[25]: Index(['Mthly\_HH\_Income', 'Mthly\_HH\_Expense', 'No\_of\_Fly\_Members',  
'Emi\_or\_Rent\_Amt', 'Annual\_HH\_Income', 'Highest\_Qualified\_Member',  
'No\_of\_Earning\_Members'],  
dtype='object')

In [26]: print(income\_df['Mthly\_HH\_Expense'].mean())

18818.0

In [27]: print(income\_df['Mthly\_HH\_Expense'].median())

15500.0

In [28]: print(income\_df['Mthly\_HH\_Expense'].mode())

0 25000  
Name: Mthly\_HH\_Expense, dtype: int64

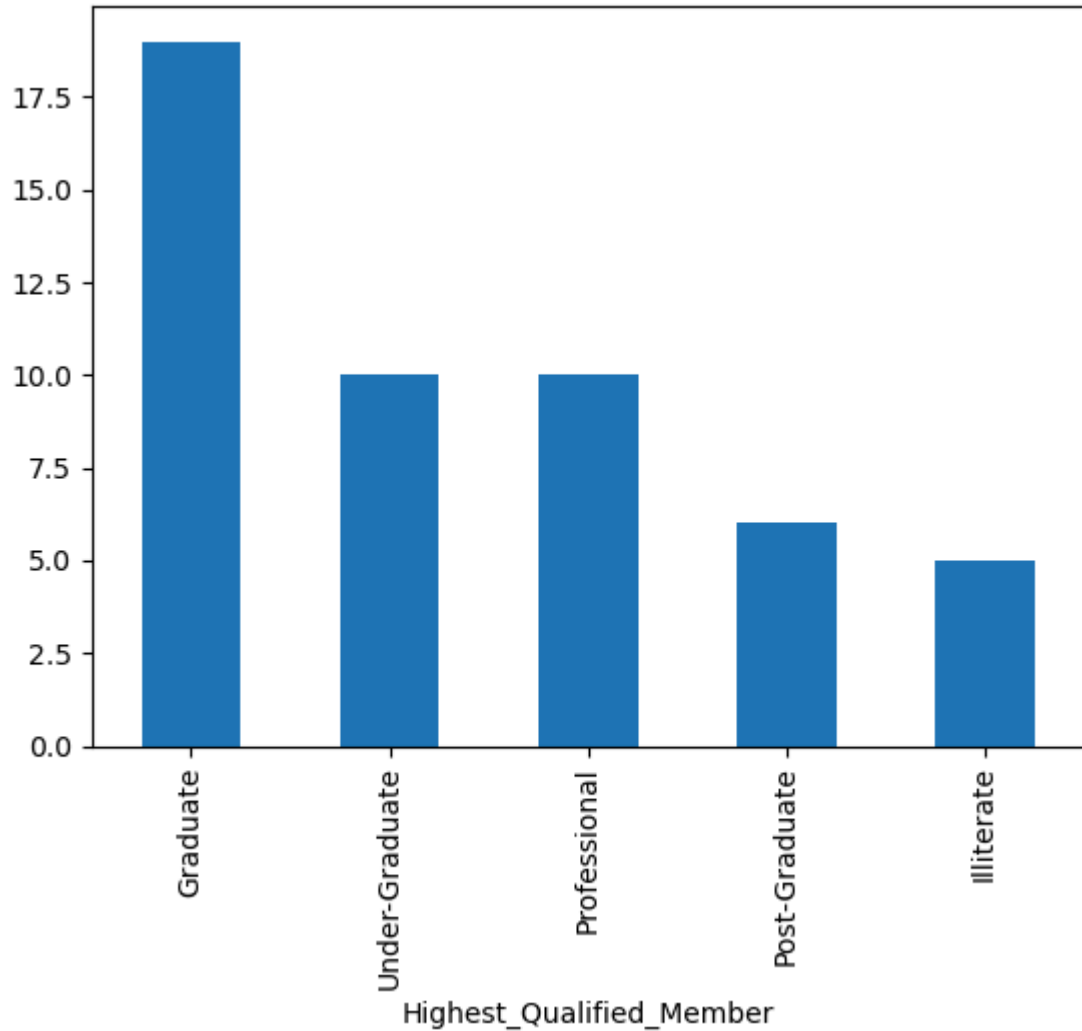
In [11]: mth\_exp\_tmp = pd.crosstab(index=income\_df["Mthly\_HH\_Expense"], columns="count")  
mth\_exp\_tmp.reset\_index(inplace=True)  
mth\_exp\_tmp[mth\_exp\_tmp['count'] == income\_df.Mthly\_HH\_Expense.value\_counts().ma

Out[11]:

| col_0 | Mthly_HH_Expense | count |
|-------|------------------|-------|
| 18    | 25000            | 8     |

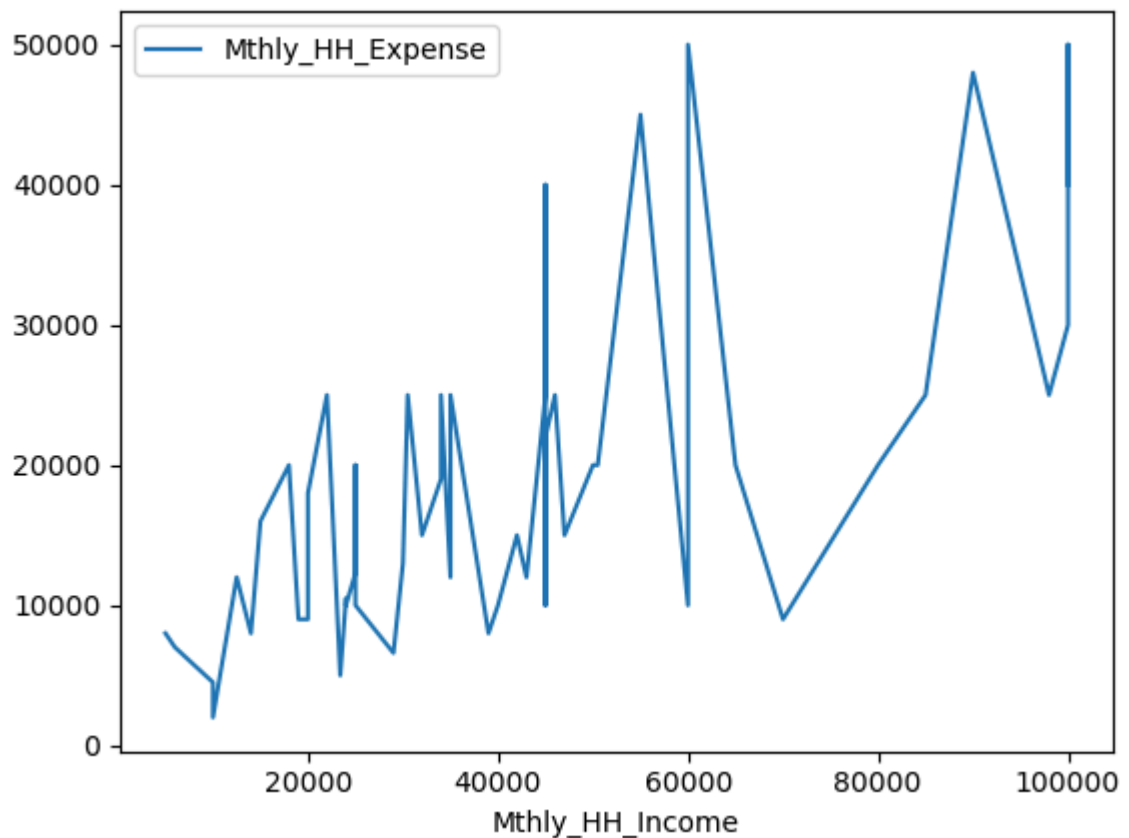
In [12]: `income_df["Highest_Qualified_Member"].value_counts().plot(kind="bar")`

Out[12]: <Axes: xlabel='Highest\_Qualified\_Member'>



In [13]: `income_df.plot(x="Mthly_HH_Income", y="Mthly_HH_Expense")`  
`IQR=income_df["Mthly_HH_Expense"].quantile(0.75)-income_df["Mthly_HH_Expense"].q`  
`IQR`

Out[13]: `np.float64(15000.0)`



## calculate standard deviation for first 4 columns

```
In [14]: pd.DataFrame(income_df.iloc[:,0:5].std().to_frame()).T
```

```
Out[14]:
```

|   | Mthly_HH_Income | Mthly_HH_Expense | No_of_Fly_Members | Emi_or_Rent_Amt | Annua |
|---|-----------------|------------------|-------------------|-----------------|-------|
| 0 | 26097.908979    | 12090.216824     | 1.517382          | 6241.434948     | 3     |

```
In [15]: # calculate variance for first 3 columns
```

```
In [16]: pd.DataFrame(income_df.iloc[:,0:4].var().to_frame()).T
```

```
Out[16]:
```

|   | Mthly_HH_Income | Mthly_HH_Expense | No_of_Fly_Members | Emi_or_Rent_Amt |
|---|-----------------|------------------|-------------------|-----------------|
| 0 | 6.811009e+08    | 1.461733e+08     | 2.302449          | 3.895551e+07    |

```
In [30]: print(income_df['Mthly_HH_Expense'].quantile(0.75))
```

25000.0

```
In [31]: print(income_df['Mthly_HH_Expense'].quantile(0.50))
```

15500.0

```
In [32]: print(income_df['Mthly_HH_Expense'].quantile(0.25))
```

10000.0

```
In [36]: IQR=income_df['Mthly_HH_Expense'].quantile(0.50)-income_df['Mthly_HH_Expense']  
IQR
```

```
Out[36]: 0      7500.0  
1      8500.0  
2     11000.0  
3     13500.0  
4      3500.0  
5      7500.0  
6      -500.0  
7     -4500.0  
8      6500.0  
9      6500.0  
10     -2500.0  
11     -9500.0  
12     10500.0  
13      5000.0  
14      5500.0  
15      3200.0  
16     -4500.0  
17      5500.0  
18      8900.0  
19      2500.0  
20     -9500.0  
21       500.0  
22     -3500.0  
23     -9500.0  
24      3500.0  
25     -9500.0  
26      7500.0  
27      5500.0  
28       500.0  
29      3500.0  
30     -9500.0  
31    -24500.0  
32      5500.0  
33     -6500.0  
34     -9500.0  
35       500.0  
36     -4500.0  
37     -4500.0  
38    -29500.0  
39      5500.0  
40    -34500.0  
41     -4500.0  
42      6500.0  
43     -4500.0  
44     -9500.0  
45    -32500.0  
46     -9500.0  
47    -14500.0  
48    -34500.0  
49    -24500.0  
Name: Mthly_HH_Expense, dtype: float64
```

```
In [37]: income_df['Mthly_HH_Expense'].var()
```

```
Out[37]: 146173342.85714287
```

```
In [38]: income_df['Mthly_HH_Expense'].std()
```

```
Out[38]: 12090.216824240286
```

```
In [40]: income_df['Annual_HH_Income'].skew
```

```
Out[40]: <bound method Series.skew of 0          64200
```

|    |         |
|----|---------|
| 1  | 79920   |
| 2  | 112800  |
| 3  | 97200   |
| 4  | 147000  |
| 5  | 196560  |
| 6  | 167400  |
| 7  | 216000  |
| 8  | 218880  |
| 9  | 220800  |
| 10 | 278400  |
| 11 | 279840  |
| 12 | 292032  |
| 13 | 316800  |
| 14 | 244800  |
| 15 | 246000  |
| 16 | 261000  |
| 17 | 258000  |
| 18 | 348000  |
| 19 | 385200  |
| 20 | 351360  |
| 21 | 445440  |
| 22 | 330480  |
| 23 | 469200  |
| 24 | 466200  |
| 25 | 449400  |
| 26 | 556920  |
| 27 | 412800  |
| 28 | 488880  |
| 29 | 619200  |
| 30 | 523800  |
| 31 | 507600  |
| 32 | 437400  |
| 33 | 610200  |
| 34 | 596160  |
| 35 | 456840  |
| 36 | 570000  |
| 37 | 581760  |
| 38 | 600600  |
| 39 | 590400  |
| 40 | 590400  |
| 41 | 647400  |
| 42 | 756000  |
| 43 | 1075200 |
| 44 | 1142400 |
| 45 | 885600  |
| 46 | 1152480 |
| 47 | 1404000 |
| 48 | 1032000 |
| 49 | 1320000 |

```
Name: Annual_HH_Income, dtype: int64>
```



```
In [34]: IQR=income_df['Mthly_HH_Expense'].quantile(0.75)-income_df['Mthly_HH_Expense']  
IQR
```

```
Out[34]: 0      17000.0  
1      18000.0  
2      20500.0  
3      23000.0  
4      13000.0  
5      17000.0  
6       9000.0  
7       5000.0  
8      16000.0  
9      16000.0  
10     7000.0  
11       0.0  
12     20000.0  
13     14500.0  
14     15000.0  
15     12700.0  
16       5000.0  
17     15000.0  
18     18400.0  
19     12000.0  
20       0.0  
21     10000.0  
22       6000.0  
23       0.0  
24     13000.0  
25       0.0  
26     17000.0  
27     15000.0  
28     10000.0  
29     13000.0  
30       0.0  
31    -15000.0  
32     15000.0  
33       3000.0  
34       0.0  
35     10000.0  
36       5000.0  
37       5000.0  
38    -20000.0  
39     15000.0  
40    -25000.0  
41       5000.0  
42     16000.0  
43       5000.0  
44       0.0  
45    -23000.0  
46       0.0  
47     -5000.0  
48    -25000.0  
49    -15000.0  
Name: Mthly_HH_Expense, dtype: float64
```

```
In [17]: income_df["Highest_Qualified_Member"].value_counts().to_frame().T
```

Out[17]:

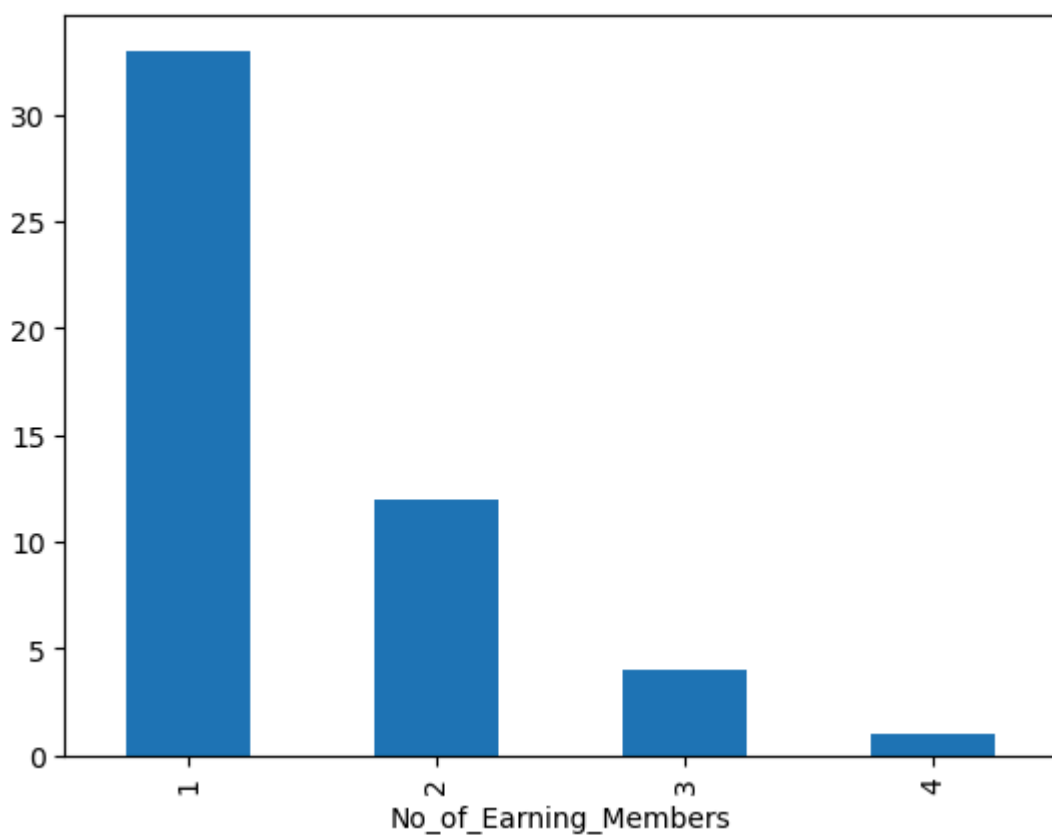
| Highest_Qualified_Member | Graduate | Under-Graduate | Professional | Post-Graduate | Illiterate |
|--------------------------|----------|----------------|--------------|---------------|------------|
| count                    | 19       | 10             | 10           | 6             | 5          |

In [ ]:

## plot the histogram to count the no\_of\_earning\_members

In [19]: `income_df["No_of_Earning_Members"].value_counts().plot(kind="bar")`

Out[19]: `<Axes: xlabel='No_of_Earning_Members'>`



In [20]: *#Here we need to calculate the coeff of variation*

```
Coeff_of_var_StockA=10/15
print(Coeff_of_var_StockA)
Coeff_of_var_StockB=5/10
print(Coeff_of_var_StockB)
```

0.6666666666666666

0.5

In [ ]: